Cue Conflict and Causal Evidence Use in Document Models on Romanian Invoices

An Early-Stage Research Proposal

Abstract

Document models can find the due date or the seller on an invoice, but it is unclear which cue they rely on: the label text, the position of the value next to the label, the reading order, the usual location of the field, or the plausibility of the value. On ordinary documents these cues agree, so accuracy cannot separate them. This project uses the cue-conflict paradigm from vision research to investigate key-value binding in documents. Synthetic Romanian invoices will set one cue against another, and each conflict will be paired with a matched control. Labels will appear in four variants: with diacritics, without diacritics, abbreviated, and in English. Four model families and a small human baseline will be compared. A second experiment will test whether the vision-language model uses the evidence box citation, by manipulating the cited value or its duplicate. No experiments have yet been conducted.

1. Introduction

Document question answering and key-information extraction ask a model to return the value of a field, such as a date or a total [1, 2]. Invoices are a demanding case because many fields share a data type and the same string can appear more than once. On a Romanian invoice, *14.05.2026* may be both the issue date (*Data emiterii*) and the delivery date. In a two-column layout, the value next to *Data scadenței* (due date) may be read as belonging to the next label. Invoice datasets [3] and Romanian financial-document studies [4] already exist, so neither the language nor the document type is new in itself.

The open question is what a model treats as evidence that a value belongs to a field. Layout models encode words and coordinates [5, 6], and vision-language models (VLMs) can answer questions and return boxes [7]. When all cues agree, however, high accuracy does not show which cue was used. Vision research solved a similar problem with stimuli in which cues disagree. ImageNet-trained networks label a cat covered in elephant skin as an elephant, whereas humans follow shape [8], and VQA-CP exposed the reliance of question-answering models on language priors [9]. I found no equivalent controlled test for key-value binding in documents. Romanian adds a testable factor: diacritics and abbreviations change how readable a label is, so we can ask whether models fall back on layout when the label becomes harder to read.

2. Related Work and Research Gap

- **Layout and shortcuts:**

Layout-aware pre-training is established [5, 6], and reading order affects extraction [10]. Several studies show that benchmark scores overstate ability. Template overlap between training and test sets inflates results [11], annotation artefacts create spurious correlations [12], and OCR-based models degrade under box, pixel and text perturbations [13]. Synthetic documents have been used to vary single factors [14]. These studies measure lost accuracy; they do not set binding cues against each other.

- **Binding:**

Language models bind entities to attributes through internal binding identifiers that can be manipulated causally [15]. A similar mechanism has been reported in VLMs for objects in synthetic scenes [16], and text-only training can encourage positional shortcuts instead of symbolic binding [17]. None of these studies concerns fields and values in documents.

- **Grounding and faithfulness:**

Correct document answers often lack correct grounding [18, 19]. In DocScope, at most 29% of correct answers had complete evidence chains [20], and when a fact repeats, a single gold region makes citation scoring unreliable [21]. Faithfulness research separates plausible from faithful explanations [22], and ERASER tests rationales by removing them [23]. In text retrieval-augmented generation, about one citation in seven was found to be attached after the answer rather than used to derive it [24]. Document benchmarks score whether a cited box *overlaps* the gold box, not whether the answer *depends* on it.

- **Gap:**

ArXiv and web searches found no study that (a) pits label text, adjacency, reading order, typical position and plausibility against each other in document key-value binding, or (b) uses box-level counterfactual edits to test whether document citations are causally used when values repeat.

3. Research Questions and Working Hypothesis

• **RQ1 (cue reliance):** When label text, adjacency, reading order, typical position and plausibility conflict, which cue determines the answer of each model family?

• **RQ2 (label legibility:** Does reliance on layout cues increase as labels become harder to read?

• **RQ3 (causal evidence use):** When a VLM cites a box, does its answer follow edits to that box and ignore edits to a non-cited duplicate? How often does overlap-based citation scoring disagree with this test?

Working hypothesis:

Models bind values to fields through a weighted combination of cues. The weights depend on the model family and on label legibility, and some citations are produced after the answer rather than used to derive it.

• **H1:** No single cue dominates. Text-only models follow label text and reading order more often; layout models and VLMs follow adjacency and typical position more often.

• **H2:** Removing diacritics or abbreviating labels increases the rate of following positional cues.

• **H3:** Under value collisions, some overlap-correct citations fail the counterfactual test.

Each hypothesis is falsifiable, and a null result is informative. If every family follows the label, a commonly assumed layout shortcut is ruled out for this setting.

4. Proposed Methodology

Original approach: No new model is proposed. The method combines a controlled stimulus generator with two protocols: cue conflicts measure *what* drives binding, and counterfactual edits measure *whether* cited evidence is used.

Generator: HTML templates will be rendered to invoice images whose word text and coordinates are known. The invoices will contain six fields: number, issue date, due date, seller, buyer and total. At least ten templates will be built, and evaluation templates will be held out from fine-tuning [11]. An *oracle-text* condition removes OCR errors as a confound.

Conflicts and labels: Each conflict item has a matched control that differs only in the manipulated cue (Table 1). Conflict items have no single correct answer: they measure which cue is followed, with human answers as the reference. Labels appear as *Data scadenței*, *Data scadentei*, *Scad.* and *Due date*. Four conflicts, four labels and the conflict/control split give 32 cells.

Table 1. Planned cue conflicts.

Conflict	Construction	The answer reveals
Label vs typical position	Issue-date and due-date blocks swap places	label or usual position
Label vs adjacency	Values shift one row, as in misaligned tables	label or nearest value
Reading order vs spatial pairing	Two columns whose linear text order pairs a label with the wrong value	spatial pair or reading order
Label vs plausibility	The due date is earlier than the issue date	label or "sensible" value

Models: F1 passes OCR text to a language model without geometry, and F2 passes the same text with verbalized coordinates. F3 is LayoutXLM [6], fine-tuned only on non-conflict invoices. F4 covers open VLMs such as Qwen2.5-VL-3B/7B [7], which return an answer and a box. Each prompt is run in two or three paraphrases.

Human baseline: Will answer about 40 conflict items and rate how realistic each invoice looks.

Counterfactual edits (F4). Each invoice is re-rendered with one minimal edit:

• **I1:** Change the value in the cited box. The answer should follow the change.

• **I2:** Change a non-cited duplicate. The answer should stay the same.

• **I3:** Change the gold box when the model cited a different copy.

• **Control:** Change an unrelated region of the same size.

5. Planned Experiments

The experiments below describe the intended direction, not a fixed protocol. Sample sizes, the models included and the evaluation details will be decided after the pilot and may change if it reveals problems.

• **Pilot.** A small set of generated invoices will be checked by me and by a few friends. The pilot should show whether the invoices look realistic, whether models answer the matched controls correctly and roughly how many documents are needed.

• **E1: How often do values repeat?** On public datasets such as SROIE, FATURA and part of DocVQA [2, 3, 1], we will estimate how often the answer to a question appears more than once on the same page. If repetitions turn out to be rare, the motivation for E3 is weaker.

• **E2: Which cue is followed? (RQ1, RQ2)** Models will answer questions on conflict invoices and their matched controls. We will compare how often each model family follows each cue, under each label variant, and how its choices relate to those of the volunteers. Conflict results are only meaningful for models that answer the controls correctly.

• **E3: Is the cited evidence used? (RQ3)** For VLMs that return a box, the edits from Section 4 will show whether the answer depends on the cited region. Of particular interest are citations that overlap the correct region but whose edit leaves the answer unchanged. The unrelated control edit indicates how much answers change by chance.

• **E4 (optional).** If time allows, an interpretability analysis such as activation patching [15, 16] could examine where binding happens inside one open VLM.

Evaluation. The main quantities are expected to be cue-following rates, agreement with human choices and, for E3, how often answers follow or ignore each edit, all reported with uncertainty estimates. Invoices generated from the same template are not independent, so the analysis will need to take the template into account. The exact statistical tests, the criterion for an overlap-correct box and the final sample sizes remain open.

6. Possible Contribution and Limitations

If the experiments succeed, the project would contribute:

1. A reusable cue-conflict protocol and generator for document key-value binding, with a human reference.
2. Cue-reliance profiles across four model families and four label variants.
3. An estimate of how often overlap-based citation scoring disagrees with causal use when values repeat.

Limitations: Synthetic documents limit generalization to real invoices, and conflict documents may look unnatural; the realism ratings check this. The human sample is small, few open models are tested, and predicted boxes may be imprecise. The study will not claim that document models are generally unreliable, and it does not propose a new architecture. Its claims are limited to the cues, models and labels tested.

7. Proposed Article Outline

Table 2. Planned structure of the final article.

Section	Planned content
1. Introduction	Problem, value collisions, research questions
2. Related Work	Layout shortcuts, binding, grounding and faithfulness
3. Stimulus Generator	Templates, conflicts, label variants, human study
4. Experimental Setup	Synthetic and public data, models, prompts, human study, metrics, pilot
5. Results: Cue Reliance	E1 audit, E2 cue-following rates and regression
6. Results: Causal Evidence Use	E3 edit outcomes, overlap vs causal-use table
7. Discussion and Threats to Validity	Interpretation, naturalness, prompts, generalization
8. Conclusion	Allowed claims and future work

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